Reading a Rational Graph from Its Formula

Answer Key

Algebra 2 Rational Functions

Quick Answers

1. $(x-3)\,(x+3)$, $(x-3)\,(x+5)$
2. $\frac{x-3}{x-2}$, $\frac{5}{4}$
3. 5, -2
4. 4, -2, $-\frac{8}{3}$
5. 3
6. $\frac{1}{2}$, 3
7. 4, -4, $-\frac{1}{8}$
8. 0
9. 2, -1, $\frac{3}{4}$
10. 4
11. 2, -6
12. $\frac{8}{3}$, 4, 2

Curriculum

Level: Algebra 2 Rational Functions
HSA-APR.D.6–D.7 – *problems 1, 2, 12*
HSF-IF.A–HSF-IF.C – *problems 3, 4, 5, 6, 7, 8, 9, 10, 11*
Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

6. *If they got -0.5: divided -6 by 12 instead of by -12, so the answer came out negative*

1. Factor $f(x) = \dfrac{x^2-9}{x^2+2x-15}$ completely.

Step 1. Numerator: $x^2 - 9$ is a difference of squares, so it factors as $(x-3)(x+3)$.
Step 2. Denominator: find two numbers with product -15 and sum $+2$. Those are $+5$ and -3, so $x^2+2x-15 = (x+5)(x-3)$.
Step 3. Write the fraction with both factorisations in place.

$$\boxed{f(x) = \frac{(x-3)(x+3)}{(x+5)(x-3)}}$$

Why this matters: the shared factor $(x-3)$ is already visible, and it is the reason $x=3$ will be a hole rather than an asymptote in later problems.

2. $g(x) = \dfrac{x^2 - x - 6}{x^2 - 4}$**: lowest terms, and the hole.**

Step 1. Factor both parts: $\dfrac{(x-3)(x+2)}{(x-2)(x+2)}$.

Step 2. The factor $(x+2)$ appears top and bottom, so it cancels for every x except $x = -2$, where the original g is undefined.

$$\boxed{g(x) = \frac{x-3}{x-2}}$$

Step 3. The height of the hole is the value the simplified form takes at $x = -2$: $\dfrac{-2-3}{-2-2} = \dfrac{-5}{-4} = \dfrac{5}{4}$.

$$\boxed{\text{hole at } \left(-2,\ \tfrac{5}{4}\right)}$$

Grading note: the simplified formula alone is only half the answer — the excluded value $x = -2$ must be stated, because g and $\frac{x-3}{x-2}$ are not the same function.

3. Vertical asymptotes of $h(x) = \dfrac{x+1}{x^2 - 3x - 10}$**.**

Step 1. Factor the denominator: two numbers with product -10 and sum -3 are -5 and $+2$, so $x^2 - 3x - 10 = (x-5)(x+2)$.

Step 2. The numerator $x+1$ is not $(x-5)$ or $(x+2)$, so nothing cancels — every zero of the denominator survives as a vertical asymptote.

Step 3. Set each surviving factor to zero: $x - 5 = 0$ and $x + 2 = 0$.

$$\boxed{x = 5}$$
$$\boxed{x = -2}$$

Listen for: a student who names $x = -1$ as an asymptote. That is the zero of the NUMERATOR, which is an x-intercept, not an asymptote.

4. Intercepts of $p(x) = \dfrac{x^2 - 2x - 8}{x^2 + 4x + 3}$**.**

Step 1. x-intercepts come from the numerator: factor $x^2 - 2x - 8 = (x-4)(x+2)$, so the numerator is zero at $x = 4$ and $x = -2$.

Step 2. Check that neither value breaks the denominator: $x^2 + 4x + 3 = (x+1)(x+3)$ is zero only at $x = -1$ and $x = -3$. Both candidates survive.

$$\boxed{x = 4}$$
$$\boxed{x = -2}$$

Step 3. The y-intercept is $p(0) = \dfrac{0 - 0 - 8}{0 + 0 + 3} = \dfrac{-8}{3}$.

$$\boxed{\left(0,\ -\tfrac{8}{3}\right)}$$

Grading note: $-\frac{8}{3} \approx -2.67$; a decimal answer is fine if the fraction is shown first.

5. Horizontal asymptote of $q(x) = \dfrac{3x^2 - 5}{x^2 + 4}$**.**

Step 1. Numerator and denominator both have degree 2, so neither grows faster — the graph settles onto the ratio of the leading coefficients rather than onto $y = 0$.

Step 2. Divide top and bottom by x^2: $\dfrac{3 - 5/x^2}{1 + 4/x^2}$. As x grows without bound, $5/x^2 \to 0$ and $4/x^2 \to 0$, leaving $\dfrac{3}{1}$.

$$\boxed{y = 3}$$

Teaching note: the graph is allowed to sit below the line for small x — a horizontal asymptote describes the far left and far right only.

6. Dana's intercepts of $r(x) = \dfrac{2x - 6}{x^2 - x - 12}$**.**

Step 1. Recompute $r(0)$ carefully. Numerator at $x = 0$: -6. Denominator at $x = 0$: -12. The quotient of two negatives is positive.

Step 2. $r(0) = \dfrac{-6}{-12} = \dfrac{1}{2}$, so Dana's mistake was dropping the denominator's minus sign.

$$\boxed{\left(0,\ \tfrac{1}{2}\right)}$$

Step 3. For the x-intercept, set the numerator to zero: $2x - 6 = 0$, so $x = 3$. The denominator at $x = 3$ is $9 - 3 - 12 = -6 \neq 0$, so the point really is on the graph.

$$\boxed{(3,\ 0)}$$

What Dana did wrong: she divided -6 by 12 instead of by -12, which flipped the sign of a correct-looking answer.

7. $s(x) = \dfrac{x + 4}{x^2 - 16}$**: asymptote or hole?**

Step 1. s is undefined wherever the denominator is zero: $x^2 - 16 = (x - 4)(x + 4) = 0$ gives $x = 4$ and $x = -4$.

$$\boxed{x = 4}$$
$$\boxed{x = -4}$$

Step 2. Cancel: $\dfrac{x + 4}{(x - 4)(x + 4)} = \dfrac{1}{x - 4}$ for $x \neq -4$. The factor $(x - 4)$ is still in the denominator, so $x = 4$ is the vertical asymptote; $(x + 4)$ cancelled, so $x = -4$ is a hole.

Step 3. The height of the hole is the simplified value at $x = -4$: $\dfrac{1}{-4 - 4} = -\dfrac{1}{8}$.

$$\boxed{\text{hole at } \left(-4,\ -\tfrac{1}{8}\right)}$$

Listen for: "two asymptotes" — the standard error. Cancelling first is what tells the two cases apart.

8. Horizontal asymptote of $t(x) = \dfrac{2x+7}{x^2+3x}$**.**

Step 1. The denominator has degree 2 and the numerator degree 1, so the bottom grows faster; a fraction with a fixed-size top and a runaway bottom is heading for zero.

Step 2. Divide through by x^2: $\dfrac{2/x + 7/x^2}{1 + 3/x}$. Every term with an x underneath tends to 0, so the whole quotient tends to 0.

$\boxed{y = 0}$

In words: far to the right the graph flattens onto the x-axis from above, getting arbitrarily close without reaching it.

9. $u(x) = \dfrac{x^2 - x - 2}{x^2 - 4}$**: which numerator zero is an intercept?**

Step 1. Factor everything: $\dfrac{(x-2)(x+1)}{(x-2)(x+2)}$. The numerator is zero at $x = 2$ and $x = -1$.

$\boxed{x = 2}$
$\boxed{x = -1}$

Step 2. $(x-2)$ cancels, so $x = 2$ is not in the domain of u at all — a point cannot be an intercept if the function has no value there. Only $x = -1$ is an x-intercept.

Step 3. At the cancelled value, the graph has a hole whose height is the simplified value there: $\dfrac{2+1}{2+2} = \dfrac{3}{4}$.

$\boxed{\text{hole at } \left(2,\ \tfrac{3}{4}\right)}$

Full answer: x-intercept at $(-1, 0)$; hole at $\left(2, \frac{3}{4}\right)$.

10. Jonah's claim about $v(x) = \dfrac{x^2 - 4}{x - 2}$**.**

Step 1. Jonah's rule — "denominator zero, therefore vertical asymptote" — skips the factoring step. Factor first: $\dfrac{(x-2)(x+2)}{x-2}$.

Step 2. The $(x-2)$ cancels, leaving $v(x) = x + 2$ for every $x \neq 2$. Nothing blows up: as x approaches 2 from either side, $v(x)$ approaches $2 + 2 = 4$.

$\boxed{\text{hole at } (2,\ 4)}$

Step 3. *Why he is wrong (model answer):* a vertical asymptote needs the function to grow without bound near the value, which happens only when a factor is left in the denominator after cancelling. Here the factor cancels, so v agrees with the line $y = x + 2$ everywhere except at $x = 2$, where it simply has no value — an open circle at $(2, 4)$.

Manual review: the explanation is the graded part; the boxed point is the check.

11. $w(x) = \dfrac{2x^2+3x}{x^2-9}$**: asymptote, and crossing it.**

Step 1. Equal degrees again, so divide by x^2: $\dfrac{2+3/x}{1-9/x^2} \to \dfrac{2}{1}$ as x grows.

$$\boxed{y=2}$$

Step 2. A crossing means $w(x)=2$, so solve $\dfrac{2x^2+3x}{x^2-9} = 2$, i.e. $2x^2+3x = 2(x^2-9)$.

Step 3. The $2x^2$ cancels: $3x=-18$, so $x=-6$. Check it is in the domain: $(-6)^2-9 = 27 \neq 0$, and $w(-6) = \dfrac{72-18}{27} = 2$ ✓

$$\boxed{x=-6}$$

Teaching note: a horizontal asymptote constrains the far ends of a graph, not the middle — crossing it once is perfectly normal.

12. Challenge: full analysis of $F(x) = \dfrac{2x^2-8}{x^2-x-2}$**.**

Step 1. Factor: $\dfrac{2(x-2)(x+2)}{(x-2)(x+1)}$, so $F(x) = \dfrac{2(x+2)}{x+1}$ for $x \neq 2$.

Step 2. The cancelled factor $(x-2)$ gives a hole at $x=2$, height $\dfrac{2(2+2)}{2+1} = \dfrac{8}{3}$.

$$\boxed{\text{hole at } \left(2,\ \tfrac{8}{3}\right)}$$

Step 3. The surviving factor $(x+1)$ gives the vertical asymptote $x=-1$; the simplified numerator is zero at $x=-2$, which is the x-intercept.

$$\boxed{x=-1}$$
$$\boxed{(-2,\ 0)}$$

Step 4. y-intercept: $F(0) = \dfrac{0-8}{0-0-2} = \dfrac{-8}{-2} = 4$.

$$\boxed{(0,\ 4)}$$

Step 5. Degrees are equal in the original formula, so the horizontal asymptote is the ratio of leading coefficients, $\dfrac{2}{1}$.

$$\boxed{y=2}$$

Model sketch (manual review): two branches of $y = \dfrac{2(x+2)}{x+1}$ separated by the dashed line $x=-1$, both approaching the dashed line $y=2$ far left and far right, crossing the axes at $(-2,0)$ and $(0,4)$, with an open circle at $\left(2, \frac{8}{3}\right)$.